

APM 1110 FA 3

Cainglet

<https://github.com/Endermite50/APM1110-CAINGLET-FA-SA>

Problem 5: Overall Percentage of Relevant Images

A geospatial analysis system has four sensors. Each sensor supplies a certain percentage of the total images, and within each sensor's images, a certain percentage are relevant to the query. The overall percentage of relevant images is a weighted average, weighted by the proportion of images each sensor supplies.

```
# Percentage of images supplied by each sensor (as proportions)
supplied <- c(0.15, 0.20, 0.25, 0.40)

# Percentage of relevant images from each sensor (as proportions)
relevant <- c(0.50, 0.60, 0.80, 0.85)
```

Overall percentage of relevant images = weighted sum:

```
overall_relevant <- sum(supplied * relevant)
overall_relevant      # as a proportion
```

```
## [1] 0.735
```

```
overall_relevant * 100      # as a percentage
```

```
## [1] 73.5
```

Conclusion: The overall percentage of relevant images is **73.5%**.

Problem 6: Pairwise vs. Mutual Independence

Sample space of two coin tosses:

```
S <- c("HH", "HT", "TH", "TT")
n <- length(S) # total outcomes = 4
```

Define the events as subsets of S:

```
E1 <- c("HH", "TT") # both tosses same
E2 <- c("HH", "HT") # first toss is head
E3 <- c("TH", "HH") # second toss is head
```

Probability function: count outcomes in event / total outcomes:

```
P <- function(event) length(event) / n
```

Intersection probability function:

```
P_and <- function(e1, e2) length(intersect(e1, e2)) / n
```

Individual probabilities:

```
P(E1); P(E2); P(E3)
```

```
## [1] 0.5
```

```
## [1] 0.5
```

```
## [1] 0.5
```

Check pairwise independence:

```
cat("P(E1)*P(E2) =", P(E1)*P(E2), " | P(E1 n E2) =", P_and(E1, E2), "\n")
```

```
## P(E1)*P(E2) = 0.25 | P(E1 n E2) = 0.25
```

```
cat("P(E1)*P(E3) =", P(E1)*P(E3), " | P(E1 n E3) =", P_and(E1, E3), "\n")
```

```
## P(E1)*P(E3) = 0.25 | P(E1 n E3) = 0.25
```

```
cat("P(E2)*P(E3) =", P(E2)*P(E3), " | P(E2 n E3) =", P_and(E2, E3), "\n")
```

```
## P(E2)*P(E3) = 0.25 | P(E2 n E3) = 0.25
```

Check mutual independence:

```
E123 <- Reduce(intersect, list(E1, E2, E3)) # triple intersection
P_triple <- length(E123) / n
P_product <- P(E1) * P(E2) * P(E3)

cat("\nP(E1 n E2 n E3) =", P_triple, "\n")
```

```
##
```

```
## P(E1 n E2 n E3) = 0.25
```

```
cat("P(E1)*P(E2)*P(E3) =", P_product, "\n")
```

```
## P(E1)*P(E2)*P(E3) = 0.125
```

```
cat("Mutually independent?", isTRUE(all.equal(P_triple, P_product)), "\n")
```

```
## Mutually independent? FALSE
```

Conclusion: Since $P(E_1 \cap E_2 \cap E_3) = 0.25 \neq P(E_1)P(E_2)P(E_3) = 0.125$, the events E_1 , E_2 , and E_3 are **pairwise independent** (every pair satisfies the multiplication rule) but **not mutually independent**, because all three share the common outcome HH , which inflates the triple intersection beyond what independence would predict.